

# Methotrexate

Methotrexate is a chemotherapy drug. It is used to treat many different cancers.

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## What is methotrexate?

Methotrexate is a type of chemotherapy. It is used to treat many different cancer types.

It is best to read this information with our general information about [chemotherapy](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

## More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

## How methotrexate is given

Methotrexate can be given in the following ways:

- a slow injection or drip (infusion) into a vein
- as tablets or a liquid to swallow
- by injection into the fluid around the spinal cord (intrathecal chemotherapy)
- as an injection into a muscle (intramuscular injection)

- by injection into an artery (inter-arterial injection), but this is rare.

You usually have methotrexate in a chemotherapy day unit or clinic as an outpatient, or during a stay in hospital. You may have it with other cancer drugs.

During your course of treatment, you will meet with someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before or on the day of each treatment, you will have a blood test. This is to check that it is safe for you to have chemotherapy.

You will meet with a doctor, nurse or pharmacist before you have chemotherapy. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your chemotherapy.

Your nurse will usually give you anti-sickness drugs before chemotherapy.

If you are having methotrexate into a vein, your nurse will give it to you as a slow injection or drip through 1 of the following:

- a [cannula](#) – a short, thin tube the nurse puts into a vein in your arm or hand
- a [central line](#) – a fine tube that goes under the skin of your chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in your arm and goes up into a vein in your chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on your chest or arm and goes into a vein in your chest.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

## Folinic acid rescue

A drug called [folinic acid](#) (also called leucovorin or calcium folinate) is sometimes given after methotrexate. It is mainly used with higher doses of methotrexate to help reduce the side effects. This is sometimes called folinic acid rescue or leucovorin rescue.

Folinic acid can be given into your cannula or line. You have it regularly through a drip until the methotrexate is out of your system. You will have fluids through the drip at the same time.

Folinic acid is sometimes given as tablets. It is very important to take the tablets exactly as explained.

## Taking your methotrexate tablets or liquid

If you are taking methotrexate as tablets or a liquid, always take it exactly as explained. This is important to make sure the treatment works as well as possible.

You should swallow the tablets whole with a full glass of water. They should not be chewed, broken or crushed.

If you have the liquid, measure out the dose carefully. Drink a glass of water after the methotrexate to rinse your mouth.

If you are sick just after taking methotrexate, contact the 24 hour number for advice. Do not take another dose. If you forget to take methotrexate, take it as soon as you remember if it is the same day. Contact your cancer team for advice. Do not take a double dose.

Other things to remember about this drug are:

- Wash your hands after taking your tablet or liquid
- Other people should avoid direct contact with the drug.
- Keep it in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep it safe, where children cannot see or reach it
- Do not throw away unused methotrexate. Return it to your cancer team at the hospital.

## Lactose

Methotrexate tablets contain a small amount of lactose. If you have a lactose intolerance, talk to your doctor before taking methotrexate tablets.

## Methotrexate by injection into the spinal fluid

In some cancers, such as some types of leukaemia or lymphoma, cancer cells can pass into the cerebrospinal fluid (CSF). You may be given [intrathecal chemotherapy](#) if a cancer has spread to the CSF, or there is a risk it could.

Only certain chemotherapy drugs, including methotrexate, can be given in this way. You will have a [lumbar puncture](#) if you need to have methotrexate by injection into the spinal fluid.

You need to lie flat for at least 30 minutes after the lumbar puncture. Your doctor or nurse will tell you how long you should stay like this for. They will check your blood pressure and pulse during this time.

A lumbar puncture is usually a very safe procedure. But you may have a headache for a few days afterwards. If this happens, tell your doctor. They can give you painkillers to help. Drinking plenty of fluids (3 litres a day) for 2 days afterwards can help to reduce headaches. If the headache does not improve, call the 24 hour number for advice.

## Your course of chemotherapy

You usually have a course of several cycles of treatment over a few months. Your cancer team will discuss your treatment plan with you.

## About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

The side effects may depend on the dose of methotrexate you are having, and how you are having it.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

## **Serious and life-threatening side effects**

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

## **Contact the hospital**

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

## **Side effects while methotrexate is being given**

Some people may have side effects while they are having a cancer drug or shortly after they have it:

### **Allergic reaction**

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

**Always call 999 if swelling happens suddenly or you are struggling to breathe.**

## The drug leaks outside the vein

Sometimes cancer drugs that are given into a vein may leak outside the vein. If this happens, some drugs can damage the tissue near the vein. This is called extravasation. Extravasation is not common, but it is important that it is dealt with quickly. If you have any of the following symptoms around the vein during or after your treatment, tell your nurse straight away:

- stinging
- pain
- swelling
- if you have white skin, the area may become red
- if you have black or brown skin, the area may become darker.

## Possible side effects

Side effects of methotrexate may be different depending on how you are having it. They also depend on the dose you are having. Your cancer team will explain what to expect.

## Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea

- needing to pass urine (pee) often, or discomfort when you pass urine.

**It is important to follow any specific advice your cancer treatment team gives you.**

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

## **Bruising and bleeding**

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

## **Feeling sick**

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

## **Tummy pain**

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

## **Sore mouth and throat**

This treatment may cause a sore mouth and throat. You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
-

your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

## **Loss of appetite**

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

## **Effects on the kidneys**

Methotrexate can affect how your kidneys work, but usually only when it is given in high doses. This will go back to normal when your treatment finishes. You will have blood tests to check how well your kidneys are working.

Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

## **Other side effects**

These side effects are less common. Some of them are much rare. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1 %).

## **Effects on the nervous system**

High doses of this treatment can affect the nervous system. You may:

- have headaches
- feel dizzy
- feel drowsy or confused.

Rarely, this treatment can cause seizures (fits). Your doctor or nurse will explain more about this.

If you have any of these symptoms, it is important to either:

- contact the hospital straight away on the 24-hour number

- go to the hospital straight away.

You should not drive yourself to hospital.

## Diarrhoea

This treatment may cause severe [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. If you have a stoma, it will be more active than usual.

Your doctor, nurse or pharmacist may give you anti-diarrhoea drugs to take at home.

If you have diarrhoea or a mild increase in stoma activity, follow their advice about:

- taking anti-diarrhoea drugs
- how much and what type of fluids to drink
- any changes to your diet that might help.

Contact the hospital straight away on the 24-hour number if any of the following things happen:

- you have diarrhoea at night
- you have uncomfortable stomach cramps
- you have diarrhoea 4 or more times in a day
- you have a moderate increase in stoma activity
- the anti-diarrhoea drugs do not work within 24 hours.

You may need to give a stool sample, or go to hospital to have fluids through a drip or antibiotics.

## Effects on the lungs

This treatment can cause inflammation of the lungs. This is called pneumonitis. Contact the hospital straight away on the 24-hour number if you notice any of these changes during treatment or after it ends:

- breathlessness
- a cough that does not go away
- wheezing
- a fever, with a temperature over 37.5°C.

You should also tell them if any existing breathing problems get worse. You may have:

- tests to check your lungs
- treatment, if needed.

## Effects on the liver

This treatment may affect how your liver works. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your liver is working.

## Hair loss

[Hair loss](#) is rare with standard doses of methotrexate. If you are having high-dose treatment with methotrexate, you usually lose all the hair from your head. You may also lose your eyelashes and eyebrows, as well as other body hair.

[Scalp cooling](#) is a way of lowering the temperature of your scalp to help reduce hair loss. Your nurse can tell you if this is an option for you.

Hair loss is almost always temporary. Your hair will usually grow back after treatment finishes.

## Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

## Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

## Skin changes

Chemotherapy can affect your skin. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

This treatment can also:

- cause a rash, which may be itchy
- make your skin darker in some areas
- make any area treated with radiotherapy become red or sore. If you have white skin the area will become red and if you have black or brown skin the area might become darker.
- make you more sensitive to the sun.

Your [skin may burn more easily](#) during treatment and for several months after. Use a sun cream of at least SPF 30. SPF stands for sun protection factor. Cover up with clothing and wear a hat.

Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help. Changes to your skin are usually temporary and improve when treatment ends.

## Serious skin changes

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

## Effects on the eyes

Your eyes may become watery and feel sore. Your doctor, nurse or pharmacist can give you advice and eye drops to help with this.

Contact the hospital on the 24-hour number if:

- your eyes get red and inflamed (conjunctivitis)
- you have pain in your eyes
- you notice any change in your vision.

## Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

## Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

## Hepatitis B reactivation

If you have had a liver infection called hepatitis B, this treatment can make it active again. Your doctor, nurse or pharmacist will talk to you about this. You may have tests for hepatitis B before, during, and after treatment.

## Second cancer

This treatment can increase the risk of developing lymphoma. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor can talk to you about this.

## Side effects of intrathecal chemotherapy

If you have [intrathecal chemotherapy](#), the side effects you have may be different from the ones listed above. Intrathecal methotrexate can cause:

- headaches
- dizziness
- tiredness
- blurred vision
- loss of balance.

These side effects can last for a few hours. If you have any of these, tell your doctor, nurse or pharmacist.

## Other information

### Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

### Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online

-

- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

## Alcohol

Avoid drinking alcohol while having this treatment. It may affect how the treatment works. It may also make some side effects worse – for example, nausea.

## Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix<sup>®</sup> for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

## Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex during this treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

## Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

## Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

## Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

## Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

## About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

### References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

## The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
-

use illustrations to explain text

- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

## How we can help

### How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

### Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



#### Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



#### Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

#### Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



#### Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



#### Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

**Get help**